Content-Based Vocal Repertoire Ranking Framework Using Duration-Weighted Pitch Distributions

Abstract

Choosing vocal repertoire that fits a singer's voice is important for vocal health, yet to the author's knowledge, current tools based on tessituragrams help assess a piece only after it has been selected—they do not recommend new pieces likely to suit the singer. A key factor in vocal fit is tessitura: the pitches on which the voice spends the most time in a given piece, weighted by how long each pitch is sustained. This paper presents a proof-of-concept content-based ranking framework that uses duration-weighted tessituragrams—pitch-by-duration profiles extracted from machine-readable musical scores—to rank songs by how well they match a singer's stated preferences. The singer specifies a comfortable vocal range, favorite pitches, and pitches to avoid; the system filters songs by range, constructs an ideal pitch profile from these preferences, and scores each candidate using a similarity measure (cosine similarity) that captures how closely the song's pitch profile aligns with the ideal, minus a penalty for time spent on pitches the singer wants to avoid. The framework is evaluated offline on a library of 101 nineteenth-century art songs from the OpenScore Lieder Corpus, using automatically generated user profiles and standard ranking metrics. The evaluation focuses on the internal properties of the ranking method, including self-retrieval accuracy, ranking stability under small preference changes, and verification that the implemented scoring formula behaves as intended. The system correctly identifies the source song as its top ranked result in 84% of test cases, outperforming a random baseline (6%), and rankings remain stable under small perturbations of user preferences. An avoid-penalty component does not improve over similarity alone in this small, homogeneous library—likely because automatically derived pitches to avoid carry limited discriminative signal. These results therefore demonstrate internal consistency and feasibility of the ranking framework, while evaluation with real singers and larger, more diverse repertoires remains future work.

Keywords: tessitura, tessituragrams, vocal repertoire recommendation, content-based recommendation, cosine similarity, music information retrieval, MusicXML

Introduction

Choosing repertoire that matches a singer's vocal characteristics is important for vocal health: misalignment between a piece's demands and a singer's capabilities can increase the risk of strain or injury (Apfelbach, 2022; Phyland et al., 1999). A central factor in those demands is tessitura—where in the vocal range the line sits most of the time, i.e. the pitches on which the voice spends the most time (Apfelbach, 2022; Schloneger et al., 2024; Thurmer, 1988). In practice, singers and teachers often filter by a piece's written range (the highest and lowest notes in the score) or by vocal classification such as Fach—the system used in opera to categorize voices by range, weight, and color. A piece's written range alone, however, does not show how much time the voice spends in different parts of that span, and Fach labels are not consistently defined across pedagogical contexts (Schloneger et al., 2024).

Tessituragrams—introduced by Thurmer in 1988—address the first limitation by showing how much time the voice spends on each pitch (Thurmer, 1988). Later work weighted pitches by rhythmic duration, so sustained notes count more than short ones (Titze, 2008). Tools such as Tessa extract tessitura from MusicXML (a machine-readable score format) and produce pitch histograms and statistics for a given piece (Apfelbach, 2022). Such tools are analytic: they help assess a piece after it is chosen, not find pieces whose tessitura is likely to fit the singer.

This paper explores whether tessituragrams can support content-based ranking of repertoire candidates based on singer preferences. We build a tessituragram-based framework that filters songs by the user's range, constructs an ideal pitch profile from favorite pitches and pitches to avoid, and ranks candidates by cosine similarity to that profile minus a penalty for time spent on avoid pitches. The system is evaluated as a proof-of-concept using an offline experiment with automatically generated user profiles to examine internal properties of the ranking method, including self-retrieval accuracy, ranking stability under small preference changes, and verification of the scoring formula.

The contributions of this research are as follows:

1. A proof-of-concept ranking pipeline combining duration-weighted tessituragrams with user-specified favorite pitches and pitches to avoid, scored using cosine similarity minus an avoid-penalty.

2. An offline evaluation framework examining internal properties of the ranking method—including self-retrieval accuracy, ranking stability under small preference changes, and formula verification—with comparison to null and cosine-only baselines.

Related Work

Tessituragram Analysis and Repertoire Selection

Thurmer (1988) formalized the tessituragram as the distribution of pitch occurrence in the vocal line; Titze (2008) added duration weighting. Recent work uses duration-based and dose-based summaries to characterize demand across works or cycles (Schloneger et al., 2024; Patinka, 2024). On the applied side, Nix (2014) evaluates objective methods—including voice range profiles, tessituragrams, and dosimetry—for matching pieces to singers in pedagogy; Apfelbach's Tessa (2022) automates tessitura extraction from MusicXML. These contributions support evaluating suitability of a chosen piece, but not recommending pieces whose tessitura is likely to fit.

Music Information Retrieval

MIR addresses content-based search and retrieval of music, including symbolic (score-based) methods (Casey et al., 2008; Gurjar and Moon, 2018). For symbolic music, pitch histograms and duration distributions are common features (Corrêa and Rodrigues, 2016). Cosine similarity suits such pitch-indexed vectors because it measures proportional alignment: two songs with the same pattern of pitch usage are considered similar regardless of total duration (Müller, 2015). For the voice, octave matters: unlike

transposition-invariant melodic similarity (Mongeau and Sankoff, 1990), tessitura modelling must keep pitch in specific octaves to reflect register and demand. Our system is related to the query-by-example paradigm in MIR (retrieving items similar to a user-provided example), though the user supplies pitch preferences rather than an example song; these preferences are matched against score-derived tessituragrams (Casey et al., 2008; Müller, 2015).

Recommender Evaluation

Offline evaluation is standard when human relevance judgments are unavailable (Herlocker et al. 2004; Urbano et al. 2013). In music recommendation, Gruson et al. (2019) showed that offline metrics can align with A/B outcomes. Shared implementations of evaluation metrics improve comparability in MIR (Raffel et al. 2014). Common ranking metrics include hit rate at k and mean reciprocal rank; Kendall's τ is standard for comparing two ranked lists (Kendall, 1948). Uncertainty is often quantified with bootstrap confidence intervals (Efron and Tibshirani, 1993). We are not aware of prior evaluation of tessitura-based vocal recommenders with these metrics and explicit baselines. Our evaluation follows these practices and is described in the following section.

Method

Data and Song Library

All MusicXML source files come from the OpenScore Lieder Corpus, an openly licensed (CC0) collection of nineteenth-century art songs with structured metadata for MIR research (Gotham and Jonas, 2022). The corpus is predominantly Lieder (German art song for voice and piano) but also includes French mélodies (the French counterpart of the art-song tradition). Both repertoires center on voice and piano, and tessitura and vocal fit are central concerns in each; we chose the corpus for the availability of high-quality MusicXML and prior MIR use. The library is limited to this art-song repertoire; generalization to other genres or eras would require further data (see Limitations).

MusicXML is a machine-readable format for notated music (notes, durations, parts, metadata). We parse it with music21 (Cuthbert and Ariza 2010), extract the vocal line only, and build a tessituragram per song: a mapping from each pitch to total duration the voice spends on that pitch. Pitches are represented as MIDI numbers (e.g. 60 = middle C). Duration is measured in quarter-note beats (one beat = one quarter note). A note held four beats contributes four times as much to the tessituragram as a one-beat note; this duration weighting reflects that sustaining a pitch is more demanding than a brief note, so the tessituragram approximates sustained vocal load (Titze 2008). Table 1 lists features extracted from MusicXML or derived and stored per song. The library contains 101 art songs (e.g. Schubert, Clara and Robert Schumann, Debussy, Fauré). Evaluation uses only this library and synthetic profiles; no human relevance judgments were collected.

Feature	Source	Description
Pitch, duration (per note)	MusicXML	Converted to MIDI number; duration in quarter beats, summed per pitch for tessituragram.
Part/voice	MusicXML	Vocal line only.
Tessituragram	Derived	MIDI pitch → total duration (duration-weighted).
min_midi, max_midi	Derived	Written pitch range of the vocal part. The lowest pitch in the range is min_midi, and the highest pitch in the range is max_midi.
Composer, title, filename	MusicXML/corpus	Work and file identification.

Table 1. Features Extracted from MusicXML and Stored per Song

Ranking Framework and Scoring

The system takes three inputs: (1) the singer's vocal range (lowest and highest comfortable note, as MIDI numbers); (2) favorite notes (pitches the singer wants maximized); (3) notes to avoid (pitches the singer wants minimized). The system proceeds in three steps.

Range filter. Songs whose written pitch range extends beyond the singer's range are excluded, because notes outside the singer's comfortable range would be difficult or impossible to sing. Only songs that fit entirely within the range are retained (candidate set C).

Ideal vector construction. A dense vector (an array with one entry for every pitch in the singer's range) is built over the pitch space [min_midi, ..., max_midi], initialized to a base weight of 0.2, with a +1.0 boost at favorite-note positions and a −1.0 penalty at notes-to-avoid positions. The base weight (0.2) is set so that favorite and avoid notes dominate the direction of the ideal vector while other pitches in range still contribute slightly; sensitivity to this value was not varied in our experiments. Any resulting negative entries are clamped to 0, ensuring the vector is entirely non-negative. The vector is then L2-normalized (divided by its Euclidean length to produce a unit-length direction vector for cosine similarity) so that the similarity depends only on the alignment of pitch distributions (direction), not on the magnitude of the ideal vector, and so that scores are comparable across singers with different range sizes (Müller, 2015).

Song scoring and ranking. Each song's tessituragram is converted to a dense vector over the same pitch space and L1-normalised (divided by its sum so that entries represent proportions of total vocal duration per pitch). This ensures that entries match the definition of tessitura (proportion of time per pitch) and allow the avoid penalty to be interpreted as the proportion of singing time on avoided pitches. Because both the song vector (durations, non-negative by definition) and the ideal vector (clamped to non-negative as described above) contain no negative entries, cosine similarity (dot product divided by the product of L2 norms) is guaranteed to lie in [0, 1] and measures alignment with preferences. The avoid penalty is the proportion of the song's vocal duration spent on notes to avoid. The final score is:

$$final_score = cosine_similarity(song, ideal) - \alpha \times avoid_penalty$$

where α (alpha) controls the weight of the avoid penalty; $\alpha = 0.5$ in the main experiments. In the implementation, *final_score* and *cosine_similarity* are each rounded to four decimal places before ranking; ranking is by *final_score* descending, with ties broken by filename.

Synthetic User Profiles

With no human relevance labels, we follow standard offline evaluation (Herlocker et al., 2004; Urbano et al., 2013). For a chosen "profile song" we set the user's range to that song's written range and take the top 4 pitches by duration in its tessituragram as favorites and the bottom 2 as notes to avoid (disjoint). That song is then treated as the designated correct match. All experiments use random seed 42 when sampling queries or profiles. Bootstrap 95% confidence intervals (CIs) use 10,000 resamples of the unit of analysis (Efron and Tibshirani, 1993).

Results

Research Question 1 (RQ1): Self-Retrieval Accuracy

Question: When the profile is derived from one song, does the system rank that song first (or in the top 3 or 5)?

Procedure: The valid query pool consists of all songs for which at least two candidates remain after range filtering (so that there is a non-trivial ranking to evaluate). We draw 50 queries at random (95 qualified). For each we build the synthetic profile and run the recommender ($\alpha = 0.5$), record the rank of the query song, and compute over the 50 queries: hit rate at 1 (HR@1) = fraction where rank = 1; hit rate at k

(HR@k) = fraction where rank ≤ k (k = 3, 5); mean reciprocal rank (MRR) = mean of 1/rank (1 = best). We compare the full model (α = 0.5), a null baseline (range filter + random ranking), and cosine-only (α = 0) on the same 50 queries. Table 2 gives the results. Both the full and cosine-only models strongly outperform the null baseline. Cosine-only is numerically slightly higher than the full model (HR@1 = 0.88 vs. 0.84), though the confidence intervals overlap; the avoid penalty does not improve self-retrieval in this setting (see Sensitivity to α for further discussion).

Because the synthetic profile is derived from the same pitch-duration representation the scorer uses, self-retrieval is a test of internal consistency rather than external validity. The 16% of queries where the query song was not ranked first are informative: they occur when another song in the library has a sufficiently similar pitch distribution to score comparably or higher, demonstrating that the system distinguishes genuinely different profiles while recognizing true similarity.

Model	HR@1 (95% CI)	HR@3 (95% CI)	HR@5 (95% CI)	MRR (95% CI)
Null (random)	0.06 [0.00, 0.12]	0.14 [0.06, 0.24]	0.26 [0.14, 0.38]	0.18 [0.12, 0.25]
Cosine-only (α = 0)	0.88 [0.78, 0.96]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	0.93 [0.87, 0.98]
Full (α = 0.5)	0.84 [0.72, 0.94]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	0.91 [0.84, 0.96]

Table 2. Self-Retrieval Accuracy and Baseline Comparison (50 Queries, 95% Confidence Interval (CI) from Bootstrap over Queries)

Research Question 2 (RQ2): Ranking Stability

Question: When we add or remove one favorite note or note to avoid, how similar is the new ranking to the original?

Procedure: We use 5 baseline profiles (candidate set ≥ 10 songs each). For each baseline, we obtain the reference ranking R_0 and generate all one-note perturbations (add or remove one favorite note or note to avoid, within the candidate set). For each perturbation we get the new ranking R_{new} and compute Kendall's τ between (Kendall, 1948): $\tau = (n_c - n_d) / (n(n-1)/2)$, where n_c = concordant pairs (ordered the same in both lists), n_d = discordant pairs (ordered differently), and n = number of songs. (Ties do not arise because scores are broken by filename, so tau-a and tau-b are equivalent here.) τ ranges from −1 to 1; τ = 1 means identical order; values above 0.7 are generally regarded as indicating strong agreement. We report mean τ per baseline with 95% bootstrap CI over baseline means. We compare the full model, cosine-only, and null baseline (random R_o vs random R_{new}). Table 3 summarizes the results (5 baselines, 130 total perturbations across all baselines). Both the full and cosine-only models show strong stability (τ > 0.7); the null baseline hovers near τ ≈ 0 as expected. Cosine-only is again numerically slightly higher (τ = 0.87 vs. 0.85), with overlapping CIs.

Model	Mean τ (95% CI)
Null (random)	-0.04 [-0.05, -0.02]
Cosine-only (α = 0)	0.87 [0.84, 0.91]
Full (α = 0.5)	0.85 [0.81, 0.88]
MRR	0.90

Table 3. Ranking Stability and Baseline Comparison (Mean τ per Baseline, 95% CI over Baselines; 5 Baselines, 130 Total Perturbations)

Implementation Verification

As a verification step, we check three properties of the scoring pipeline: (a) Do scores spread out across songs, or are rankings driven by noise? (b) Does *final_score* = cos − α × avoid hold numerically

(implementation correctness)? (c) Do components correlate in the directions the formula predicts (sanity check)?

Procedure: We use 25 synthetic profiles (≥ 10 songs each) and compute unrounded scores per song (to verify the exact numerical identity of the formula, independent of the rounding step applied for ranking). (a) Spread: mean variance and mean range of *final_score* per profile; 95% CI over profiles. (b) Formula: Identity residual $|final - (cos - \alpha \times avoid)|$ (mean and max); regression $final \sim cos + avoid$ (expect coefficients ≈ 1 and ≈ −0.5, $R^2 \approx 1$). (c) Sanity check: Spearman's ρ (rank correlation; Spearman 1904) for final–cos, final–avoid, and cos–favorite_overlap; aggregated via Fisher z-transform (a standard technique for averaging correlation coefficients across samples of different sizes; Fisher 1915): each ρ is transformed to $z = \text{atanh}(\rho)$, a weighted mean z is computed (weights $n - 3$), and the result is back-transformed via $\rho = \tanh(\text{mean } z)$; 95% CI by bootstrap over profiles. Table 4 gives spread, correlations, and identity/regression results. Scores spread meaningfully (mean range ≈ 0.76); the identity residual is 0; regression recovers coefficients of 1 and −0.5 with $R^2 = 1$; and correlation signs match the formula.

Quantity	Value (95% CI where applicable)
Mean variance (*final_score*)	0.040 [0.035, 0.046]
Mean range (*final_score*)	0.758 [0.715, 0.798]
ρ(final, cos)	0.987 [0.982, 0.992]
ρ(final, avoid)	-0.35 [-0.51, -0.16]
ρ(cos, favorite_overlap)	0.935 [0.920, 0.946]
Identity	final – (cos – α × avoid)
Regression (coef_cos, coef_avoid, R^2)	1.0, -0.5, 1.0

Table 4. Implementation Verification: Spread, Sanity-Check Correlations, and Formula Checks (25 Profiles)

Sensitivity to α

To assess how sensitive the results are to the avoid-penalty weight, we repeated the self-retrieval and stability experiments for α ∈ {0.0, 0.25, 0.5, 0.75, 1.0} on the same queries and baselines. Table 5 shows HR@1 and MRR (self-retrieval) and mean τ per baseline (stability) for all five values.

α	HR@1 (95% CI)	MRR (95% CI)	Mean τ (95% CI)
0.0	0.88 [0.78, 0.96]	0.93 [0.87, 0.98]	0.87 [0.84, 0.91]
0.25	0.86 [0.76, 0.94]	0.92 [0.86, 0.97]	0.86 [0.83, 0.88]
0.5	0.84 [0.74, 0.94]	0.91 [0.84, 0.96]	0.85 [0.81, 0.88]
0.75	0.84 [0.72, 0.94]	0.91 [0.84, 0.96]	0.83 [0.80, 0.87]
1.0	0.84 [0.72, 0.94]	0.91 [0.84, 0.96]	0.83 [0.78, 0.86]

Table 5. Alpha Sensitivity: Self-Retrieval HR@1 and MRR; Stability Mean τ per Baseline (95% CI)

Point estimates at α = 0.5 match Tables 2 and 3; minor CI differences (e.g. [0.74, 0.94] vs. [0.72, 0.94] for HR@1) reflect independent bootstrap resampling across experiment runs and are within expected sampling variability. Metrics remain high across all α values and vary smoothly, confirming that the system is not brittle at any single operating point. However, the trend is clear: performance is highest at α = 0 (cosine-only) and decreases monotonically as α increases. The avoid penalty does not improve self-retrieval accuracy or ranking stability in this evaluation. This likely reflects the small library (101 songs) and the synthetic profile design, where the notes to avoid (the two pitches with the least total duration) contribute little to most candidate songs' tessituragrams and thus offer limited discriminative signal. The avoid

mechanism remains a principled design choice—penalizing time on notes a singer wishes to avoid is musically meaningful—but its benefit may require a larger, more diverse library or real user preferences to manifest.

Discussion and Limitations

The results confirm that range filtering combined with cosine similarity over duration-weighted pitch profiles captures meaningful differences between songs and produces stable rankings. In this evaluation, the cosine-only model performs slightly better than the full model that includes the avoid penalty: self-retrieval and stability both decline as α increases. The avoid term is conceptually motivated—it allows singers to indicate pitches they wish to avoid—but in the present dataset and with synthetically derived preferences it does not provide additional empirical benefit. Within this study, therefore, the best-performing model is the simpler cosine-only variant. Several limitations qualify these findings:

- No human judgments. Evaluation is fully offline with synthetic profiles; results indicate internal consistency and robustness, not user satisfaction.
- Self-retrieval circularity. The synthetic profile is derived from the same representation used for scoring, so the self-retrieval task tests internal consistency rather than external validity with real users.
- Small evaluation scale. The self-retrieval experiment uses 50 sampled queries (from 95 qualified); the stability experiment uses only 5 baseline profiles. Bootstrap CIs quantify sampling uncertainty, but the estimates would be more precise with a larger library.
- Single corpus. The 101-song library is drawn entirely from nineteenth-century art songs: German Lieder (Schubert, Clara Schumann, Robert Schumann) and French mélodies (Debussy, Fauré). This stylistic homogeneity means pitch distributions across songs are more similar than in a mixed-genre library, which may inflate self-retrieval failures (more near-duplicates in pitch profile), ease the stability test (small perturbations have small effects in a homogeneous space), and narrow score spread relative to what a diverse corpus would produce. Generalisation to other genres, eras, or larger libraries remains untested.
- Pitch and duration only. The system uses only pitch and duration information from the score. Other factors that affect vocal suitability—such as dynamics (loud vs. soft markings), tempo, text setting, and accompaniment difficulty—are not modelled.
- Avoid penalty does not outperform cosine-only. As discussed above, the full model ($\alpha = 0.5$) does not improve on cosine-only ($\alpha = 0$) on any metric in this evaluation. The avoid term's value may depend on richer preference data or a more heterogeneous library.

Conclusion

We presented a content-based ranking framework that uses duration-weighted tessituragrams to rank vocal repertoire by how well each song matches a singer's pitch preferences. The system filters songs by the singer's comfortable range, constructs an ideal pitch profile from favorite notes and notes to avoid, and scores candidates using cosine similarity minus a penalty for time on avoided pitches. Evaluated on a library of 101 nineteenth-century art songs using synthetic user profiles, both the full model and a cosine-only baseline strongly outperform a null baseline. Rankings are stable under small preference perturbations, and implementation verification confirms alignment between the code and the intended scoring formula.

The avoid penalty does not improve over cosine-only in this setting, likely because of the small, homogeneous library and synthetic derivation of notes to avoid limiting discriminative signal. Future work should evaluate the system with real user preferences and human relevance judgments, expand the library to include more diverse repertoire (e.g. opera arias, musical theatre, popular song), and incorporate additional vocal-suitability factors such as dynamics, tempo, and text setting. The tessituragram-based approach provides a principled and extensible foundation for vocal repertoire ranking systems.

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